ENVIRONMENTAL POLLUTION CONTROL TECHNOLOGIES	
Subject Code: CVOE04	**Credits: 3:0:0**
Pre requisites: - Nil	**Contact Hours: 42L**
Course Coordinator: Dr. A Lavanya	

Course Content

Unit I

Domestic Wastewater

Introduction, description of hydrologic cycle, various sources and types of wastewater. Management of wastewater within the building: different types of plumbing system-one pipe and two pipe systems. Conventional flow diagram of wastewater treatment. Concept of BOD and COD. BOD kinetics and problems, general standards for discharge of environmental pollutants. Working principles: Rotating biological contractor (RBC), moving bed biological reactor (MBBR).

- Pedagogy/Course delivery tools:-Chalk and talk, Power point presentation
- Links: -https://www.youtube.com/watch?v=sBnrwhYKbEg
  https://www.youtube.com/watch?v=z-59Og-CbgQ
  https://www.youtube.com/watch?v=SPUa7A5hRCc
- NPTEL Links: -https://www.youtube.com/watch?v=4-SRMmqH2s4&t=8s
  https://www.youtube.com/watch?v=TZlA-6tHqTo

Unit II

Industrial Wastewater

Introduction, **w**ater usage in Industry, Industrial wastewater treatment - Theories and Practices: Volume Reduction: classifying wastes; conserving wastewater; changing production to decrease wastes; reusing both industrial and municipal effluents as raw water supplies and eliminating batch or slug discharges of process wastes. Concentration Reduction: process changes, equipment modifications, segregation of wastes, equalization of wastes, by-product recovery and proportioning wastes. Pre-treatment of Industrial wastewater –Neutralization: Mixing of wastes and chemical treatment. Equalization and Proportioning: objectives and methods.

- Pedagogy/Course delivery tools:-Chalk and talk, Power point presentation, videos
- Links:https://youtu.be/SRvRFY-c5
  U?list=PL4SPMn8HFBYBhLHBY8D7iCNek9sWGWPCX
  https://youtu.be/QO5X0iaq2S8?list=PL4SPMn8HFBYBhLHBY8D7iCNek9sWGWPCX

- NPTELLinks:https://youtu.be/8FjhsODW1q4?list=PLLy_2iUCG87Cr__rs9sS1zSaR62imd0uB
  https://youtu.be/Ar04qG1P8Es?list=PLLy_2iUCG87Cr__rs9sS1zSaR62imd0uB

Unit III
Solid Waste Management

Introduction, classification and characteristics of solid wastes, Stages in municipal solid waste management. Composting and its methods. Landfills: factors for landfill site selection, Landfill design approach, Leachate and gas collection system. Solid waste management in apartments. Waste to energy conversion approaches.

- Pedagogy/Course delivery tools:-Chalk and talk, Power point presentation
- Links: -https://archive.nptel.ac.in/courses/105/103/105103205/
  https://www.youtube.com/watch?v=Hi4zFTqzf7Qhttps://www.youtube.com/watch?v=cjIacnNRLHE&t=10s
- NPTEL Links: -https://www.youtube.com/watch?v=ZHdBK5QDd54&t=3s
  https://www.youtube.com/watch?v=hu51ENxiR_M&t=1s

Unit IV
Air & Noise Pollution

Air & Noise pollution: Introduction, history of air pollution. Sources of air pollution and classification of air pollutants. Ambient air quality monitoring: dust fall jar, high volume air sampler and problems. Sources of noise pollution. Standards of noise recommended by Ministry of Environment and Forest. Preventive measures.

- Pedagogy/Course delivery tools:-Chalk and talk, Conventional water treatment plant visit, invited lectures from industry people, Power Point Presentation
- Links:https://www.youtube.com/watch?v=u142u-wJFwI&list=PLLy_2iUCG87BwOQUbS7WSdMVWHDXByk-w&index=2
  https://youtu.be/StL9XX7qwCc?list=PLLy_2iUCG87BwOQUbS7WSdMVWHDXByk-w
- NPTEL Links: -https://youtu.be/5dukz1UOtkA
  https://youtu.be/DAQapF-F4Vw
  https://youtu.be/I7Z34WU257U?list=PLLy_2iUCG87BwOQUbS7WSdMVWHDXByk-w

Unit V
Environmental Impact Assessment

Definition of EIA, need for EIA, EIS, FONSI, REIA, CEIA, step-by-step procedure for conducting EIA, limitations of EIA, methodologies of EIA: listing, Adhoc method,

Network method and Overlay method. Public participation in environmental decision making, objectives of public participation and public participation techniques. Practical consideration in preparing in EIA and EIS.

- Pedagogy/Course delivery tools:-Chalk and talk, invited lectures from industry people,Powerpoint Presentation, video.
- Links:https://youtu.be/8vymfIfip2w?list=PLLy_2iUCG87CkrNdXME16BCptwGx1fl67
  https://youtu.be/E_oRwSRdgcQ?list=PLLy_2iUCG87CkrNdXME16BCptwGx1fl67
  https://www.youtube.com/watch?v=HlBYCtJ9_Cw
- NPTEL Links: -https://www.youtube.com/watch?v=WIqeZH_0jqQ
  https://youtu.be/E_oRwSRdgcQ?list=PLLy_2iUCG87CkrNdXME16BCptwGx1fl67

Text Books:

1. Garg. S.K, "Sewage disposal and Air pollution engineering", Khanna Publications, 2-019, ISBN-10: 9788174092304
2. Industrial Waste Treatment Contemporary Practice and Vision for the Future by Nelson Leonard Nemerow, Elsevier, 2007, ISBN-13: 978-0-12-372493-9.
3. George Tchobanoglouset.al., "Integrated Solid Waste Management', Tata-McGraw-Hill, Inc. New York, 1993.
4. Rao, M.N and Rao, H.V.N. (1993) 'Air Pollution', Tata-McGraw-Hill Publishing Company Ltd., New Delhi, India.
5. Y. Anjaneyulu and Valli Manickam, "Environment Assessment Methodologies", B.S Publications, Hyderabad, 2007.

Reference Books:

1. Rao.C.S, (1992) "Environmental Pollution Control Engineering", Wiley Eastern Limited.
2. Gilbert M Masters, (2004), "Introduction to Environmental Engineering and Science" Second Edition. Pearson Education.
3. Mahajan.S.P, "Pollution Control in Process Industries", Tata McGraw Hill Publishing Co., New Delhi.
4. A.D.Bhide and B.B.Sudareshan, "Solid waste management in Developing countries", NEERI, Nagpur, 1983.
5. Larry W Canter, "Environmental Impact Assessment" –McGraw – Hill International Editions, 1996.
6. Pollution Control in Process Industries by S P Mahajan, McGraw Hill Education (2017), ISBN-13: 978-0074517727.

Web links and video Lectures (e- Resources):

1. https://www.youtube.com/watch?v=4-SRMmqH2s4&t=8s
2. https://www.youtube.com/watch?v=TZlA-6tHqTo
3. https://youtu.be/8FjhsODW1q4?list=PLLy_2iUCG87Cr__rs9sS1zSaR62imd0uB
4. https://www.youtube.com/watch?v=ZHdBK5QDd54&t=3s
5. https://youtu.be/I7Z34WU257U?list=PLLy_2iUCG87BwOQUbS7WSdMVWHDXByk-w
6. https://youtu.be/E_oRwSRdgcQ?list=PLLy_2iUCG87CkrNdXME16BCptwGx1fl67

Course Outcomes (COs):

At the end of the course, students will be able to

1. Identify the domestic wastewater sources and its treatment technologies. (PO - 1 & PSO - 1)
2. Identify the industrial wastewater control technologies. (PO - 1,2)
3. Identify municipal solid waste management techniques. (PO - 1,2)
4. Identify Air and noise pollution sources and control techniques. (PO - 1,2)
5. Describe the fundamental concepts of EIA and its methodologies, along with significance of public participation in EIA. (PO - 1,2).

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	**Marks**	**Course out comes mapped**
Internal Test-I	30	CO1, CO2 & CO3
Internal Test-II	30	CO4 & CO5
Average of the two internal tests shall be taken for 30 marks		
Other components		
Assignment – MCQ, Quiz, Group presentation	10	CO1, CO2, CO3, CO4 & CO5
Assignment – MCQ, Quiz, Group presentation	10	CO1, CO2, CO3, CO4 & CO5
Semester End Examination (SEE)	100	CO1, CO2, CO3, CO4, CO5